Student Performance Prediction

A Dual-Objective Machine Learning Framework for Academic Risk Detection

REGRESSION (CONTINUOUS MARKS) * CLASSIFICATION (PASS / FAIL VERDICT)

Machine Learning Project | Semester Assessment & Lab Viva Presentation

INTRODUCTION

Project Overview & Core Objectives

Objective 1: Marks Prediction (Regression)

* Task Type: Supervised Machine Learning (Continuous)
* Estimates exact percentage / final exam marks (0 - 100).
* Inputs: Daily study hours, attendance, past exam score, sleep.
* Algorithm: Multiple Linear Regression & Random Forest Regressor.
* Benefit: Gives transparent scoring benchmarks and targets for student

Objective 2: At-Risk Detection (Classification)

* Task Type: Binary Classification (Pass / Fail Verdict)
* Identifies students at risk of scoring below passing grade (<40%).
* Calculates calibrated pass probability via Sigmoid function.
* Algorithm: Logistic Regression & Random Forest Classifier.
* Benefit: Enables teachers to trigger remedial intervention 6 weeks early.

SYSTEM DESIGN

End-to-End System Pipeline & Architecture

1. Data Ingestion

- 1,000 synthetic records
- Realistic distributions
- 5 core student features
- CSV dataset format

2. Preprocessing

- Train-Test split (80/20)
- Split BEFORE scaling
- Prevents Data Leakage
- Deterministic random state

3. Feature Scaling

- StandardScaler applied
- Zero mean, unit variance
- Fitted on Train ONLY
- Transforms Test & Train

4. Dual Modeling

- Linear Regression
- Logistic Regression
- Random Forest Regressor
- Random Forest Classifier

5. Evaluation

- MAE, RMSE, R2 Score
- Accuracy & F1-Score
- Confusion Matrix check
- Model artifact serialization

DATA ENGINEERING

Feature Schema & Dataset Exploration

study_hours_per_day

Continuous (1.0 - 10.0 hrs)

Daily self-study hours outside regular lectures. Major driver of marks.

attendance_percentage

Continuous (45% - 100%)

Lecture attendance rate; reflects consistency and active classroom presence.

previous_marks

Continuous (20% - 100%)

Past semester examination percentage representing academic baseline.

practice_tests_taken

Discrete (0 - 12 tests)

Number of mock exams attempted; indicates revision and test familiarity.

sleep_hours

Continuous (4.0 - 10.0 hrs)

Average sleep duration per night; accounts for physical cognitive health.

final_marks / pass_fail

Target Variables (Continuous / Binary)

Continuous final score (0-100) and binary pass/fail class (1 if marks >= 40).

PERFORMANCE BENCHMARKS

Model Evaluation & Experimental Results

89.42%

R2 Score

Linear Regression model explains 89.4% of sco

+/- 3.00

MAE (Marks)

Average deviation from actual marks is only +/-

99.50%

Accuracy

199 out of 200 holdout students accurately

99.75%

F1-Score

Exceptional harmonic balance between Precision & Reca

Detailed Experimental Analysis & Observations:

* Generalization: Linear Regression generalized better than Random Forest on continuous marks (R2: 89.4% vs 83.7%).
* Error Bounds: Low RMSE of 3.80 marks demonstrates tightly bounded error without heavy outlier skew.
* Classification Strength: Logistic Regression achieved 100% Precision and 99.5% Recall on 200 hold-out test samples.
* Confusion Matrix: Out of 200 test cases, 199 were True Positives and 0 False Negatives were recorded.

EXPLAINABLE AI

Feature Weights & Educational Insights

Linear Regression Feature Weights

Daily Study Hours (+7.159)
Largest single positive contributor to exam marks.

Previous Marks (+5.233)
Baseline subject grasp anchors student trajectory.

Practice Tests Taken (+4.699)
Mock tests build exam temperament and pacing.

Attendance Rate (+4.033)
Direct classroom learning ensures syllabus coverage.

Sleep Hours (+0.754)
Consistent rest prevents mental exhaustion.

Actionable Academic Takeaways

1. Study Consistency: Adding 1.5 hours of daily focused study yields ~4.2 extra

2. Mock Tests Priority: Attempting 3-4 practice tests can push borderline student

3. Attendance Threshold: Students maintaining >75% attendance show negligibl

4. Timely Mentorship: Automated prediction flags struggling students 4-6 weeks

LIVE DEMONSTRATION

Inference Engine & Deployment

Live Single Student Evaluation Test Case

Student Profile Input Parameters:

* Study Hours: 5.5 hrs/day | Attendance: 82.0% | Previous Marks: 70.0% | Sleep: 7.5 hrs | Mock Tests: 4

Model Prediction Outputs:

* Predicted Final Marks: 67.12 / 100 (First Class Grade)

* Pass/Fail Verdict: PASS (Calibrated Pass Probability: 99.98%)

* Inference Response Time: < 5 milliseconds

* Deployed Interfaces: Interactive Web App (Streamlit `app.py`) & Standalone CLI (`predict.py`)

VIVA DEFENSE

Frequently Asked Viva Questions & Answers

Q1: Why apply StandardScaler AFTER train-test split?

Ans: To prevent Data Leakage. Computing mean/variance across the whole dataset

Q2: Why use Logistic Regression instead of Linear Regression for Pass/Fa

Ans: Linear Regression outputs unbounded values (<0 or >1). Logistic Regression squashes va

Q3: What does an R2 Score of 0.8942 represent?

Ans: 89.42% of the variance in final student marks is directly explained by the 5 inpu

Q4: How do you handle class imbalance in student pass/fail?

Ans: Evaluated via Precision, Recall, and F1-Score; techniques like SMOTE or class-weight adj

SUMMARY

Conclusion & Future Scope

Key Takeaways

* Unifies both continuous marks estimation and binary failure risk detection under one cohesive ML pipeline.

* Demonstrated high predictive accuracy (R2: 89.4%, Accuracy: 99.5%) with tight error bounds (+/- 3 marks MAE).

* Clear feature interpretability: Daily study hours and past exam performance identified as paramount factors.

Future Enhancements

* LMS Integration: Direct API integration with university portal (Canvas, Moodle, or custom ERP).

* Deep Learning: Multi-Layer Perceptrons (ANNs) to capture non-linear behavioral patterns and assignments.

* Automated Advisory: Trigger automated SMS and email notifications to academic advisors for at-risk students.